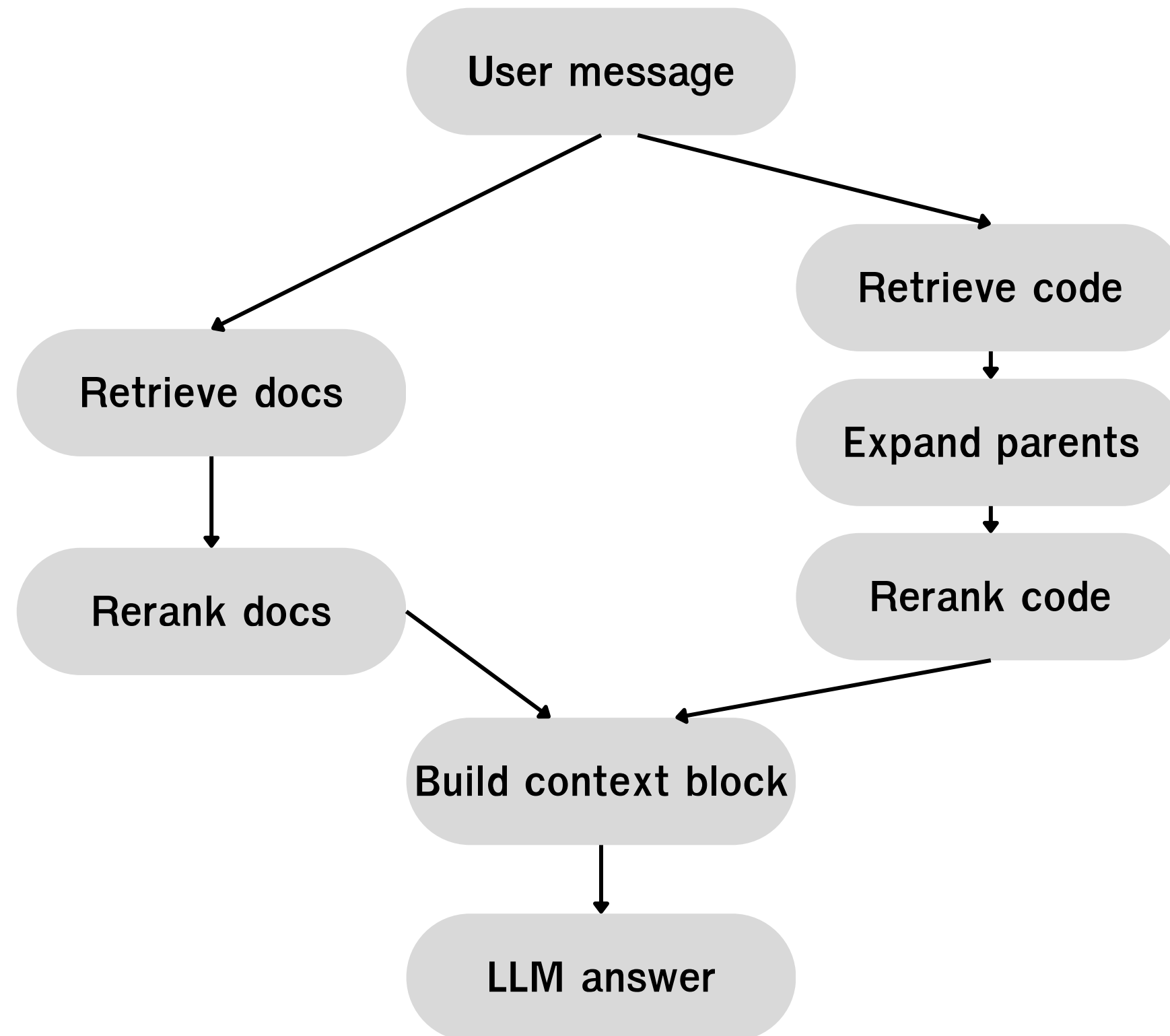




1st prompt

llm: gpt-4.1-mini

```
ANSWER_PROMPT = ChatPromptTemplate.from_template(
    """You are an expert assistant specialised in Eclipse Epsilon.

    Use the provided context as the PRIMARY and AUTHORITATIVE
    source of truth.
    Only use syntax, class names, methods, and patterns that appear
    explicitly in the context.
    If the context is insufficient for a complete answer, say so clearly.

    {context_block}

    Question:
    {question}

    Answer:"""
)
```

can run with no €	correct	complete	retrieval	faithfulness
no	no	no	no	low
no	no	no	yes	medium
no	no	no	yes	medium
no	no	no	yes	medium
yes	no	no	yes	low
yes	yes	no	yes	high
no	no	no	yes	medium
no	no	no	yes	medium
no	no	no	yes	medium
no	no	no	yes	medium

2nd prompt

llm: gpt-4.1-mini

```
ANSWER_PROMPT = ChatPromptTemplate.from_template(
    """You are an expert assistant specialised in Eclipse Epsilon.
```

Use the provided context as the PRIMARY and AUTHORITATIVE source of truth.
Only use syntax, class names, methods, and patterns that appear explicitly in the context.
You may adapt documented patterns to the user's YAML structure, provided that no new APIs or unsupported methods are invented.
If the context is insufficient for a complete answer, say so clearly.

{context_block}

Question:
{question}

Answer:"""
)

can run with no c	correct	complete	retrieval	faithfulness
no	no	no	no	low
yes	yes	no	yes	high
no	no	no	yes	medium
no	no	no	no	low
yes	yes	no	yes	high
yes	yes	no	yes	high
no	no	no	yes	medium
no	no	no	yes	medium
yes	yes	no	yes	high
no	no	no	yes	medium

1st prompt

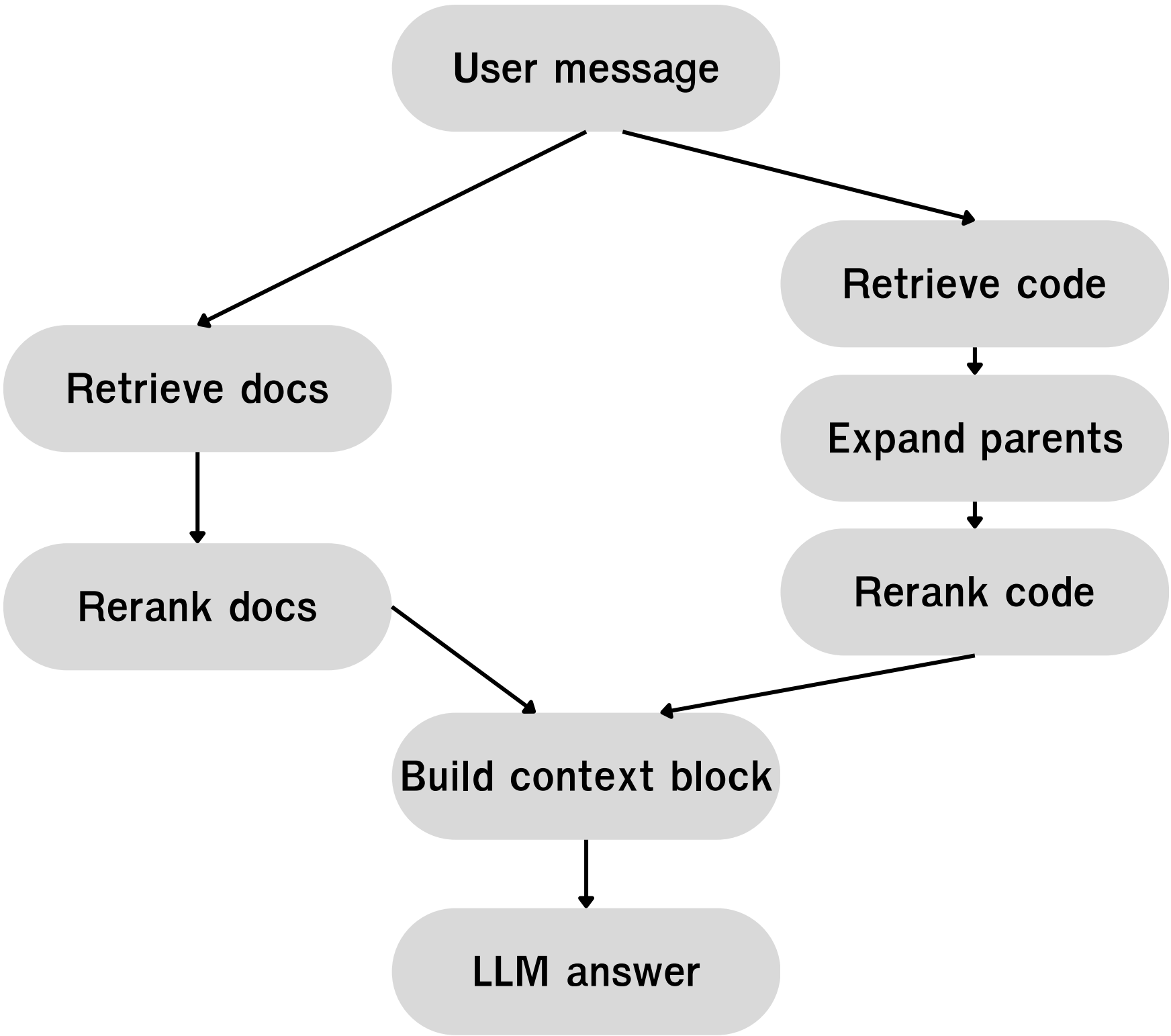
can run with no error	correct	complete	retrieval	faithfulness
no	no	no	no	low
no	no	no	yes	medium
no	no	no	yes	medium
no	no	no	yes	medium
yes	no	no	yes	low
yes	yes	no	yes	high
no	no	no	yes	medium
no	no	no	yes	medium
no	no	no	yes	medium
no	no	no	yes	medium

2nd prompt

can run with no error	correct	complete	retrieval	faithfulness
no	no	no	no	low
yes	yes	no	yes	high
no	no	no	yes	medium
no	no	no	no	low
yes	yes	no	yes	high
yes	yes	no	yes	high
no	no	no	yes	medium
no	no	no	yes	medium
yes	yes	no	yes	high
no	no	no	yes	medium

- 2nd prompt give more faithfulness, correct answer with no error code but the most of them not complete
- retrieval is important. If not find when retrieve → answer will be wrong

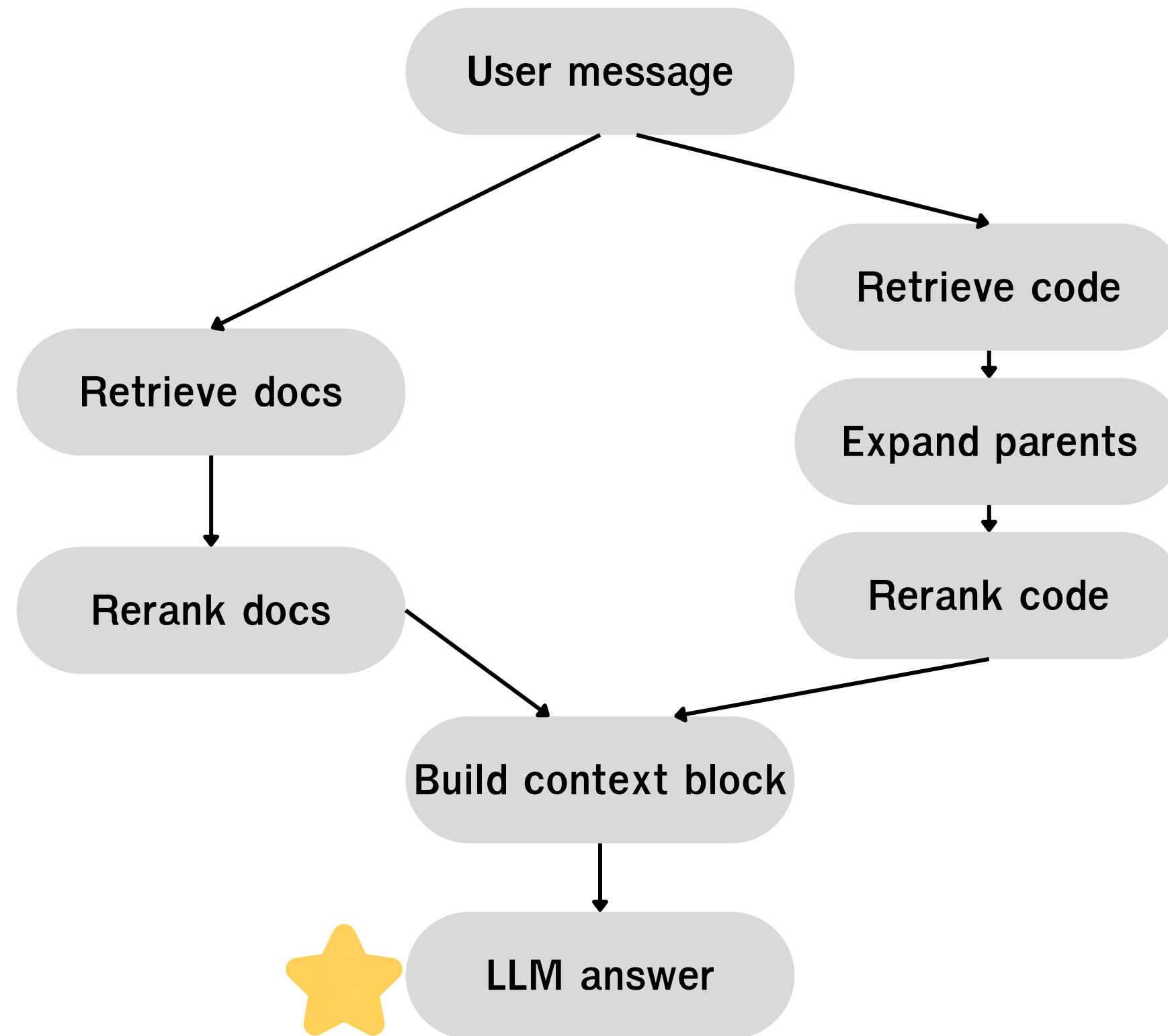
changing LLM for answer



llm: gpt-5-mini

give all full code	give example	can run with no	correct	complete	retrieval	faithfulness
no	no	no	no	no	no	low
no	yes	yes	yes	no	yes	high
yes	yes	yes	yes	yes	yes	high
no	no	no	no	no	no	low
no	yes	yes	yes	no	yes	high
no	yes	yes	yes	no	yes	high
no	yes	no	no	no	yes	medium
yes	yes	yes	yes	yes	yes	high
yes	yes	yes	no	no	yes	medium
no	yes	yes	yes	no	yes	high

- Changing LLM can help
- adapt code more correctly
 - but the important thing is if it retrieve wrong document → LLM can't answer correctly → need to focus on question only



document

"How can I create a YAML document from scratch?"

chunk vector: [create: 0.4, yaml: 0.3, node: 0.2, eol: 0.1]

"How can I create a node using YAML EMC in EOL?"

- studentId: S1024

courses:

- code: CS301

title: Machine Learning

attendance:

percentage: 92"

query vector: [yaml: 0.5, create: 0.2, studentId: 0.1, courses: 0.1, node: 0.05, eol: 0.05]

cosine similarity = $(0.4 \times 0.2) + (0.3 \times 0.5) + (0.2 \times 0.05) + (0.1 \times 0.05) + (0.1 \times 0.05)$
 $= 0.08 + 0.15 + 0.01 + 0.005 + 0.005 = 0.25$

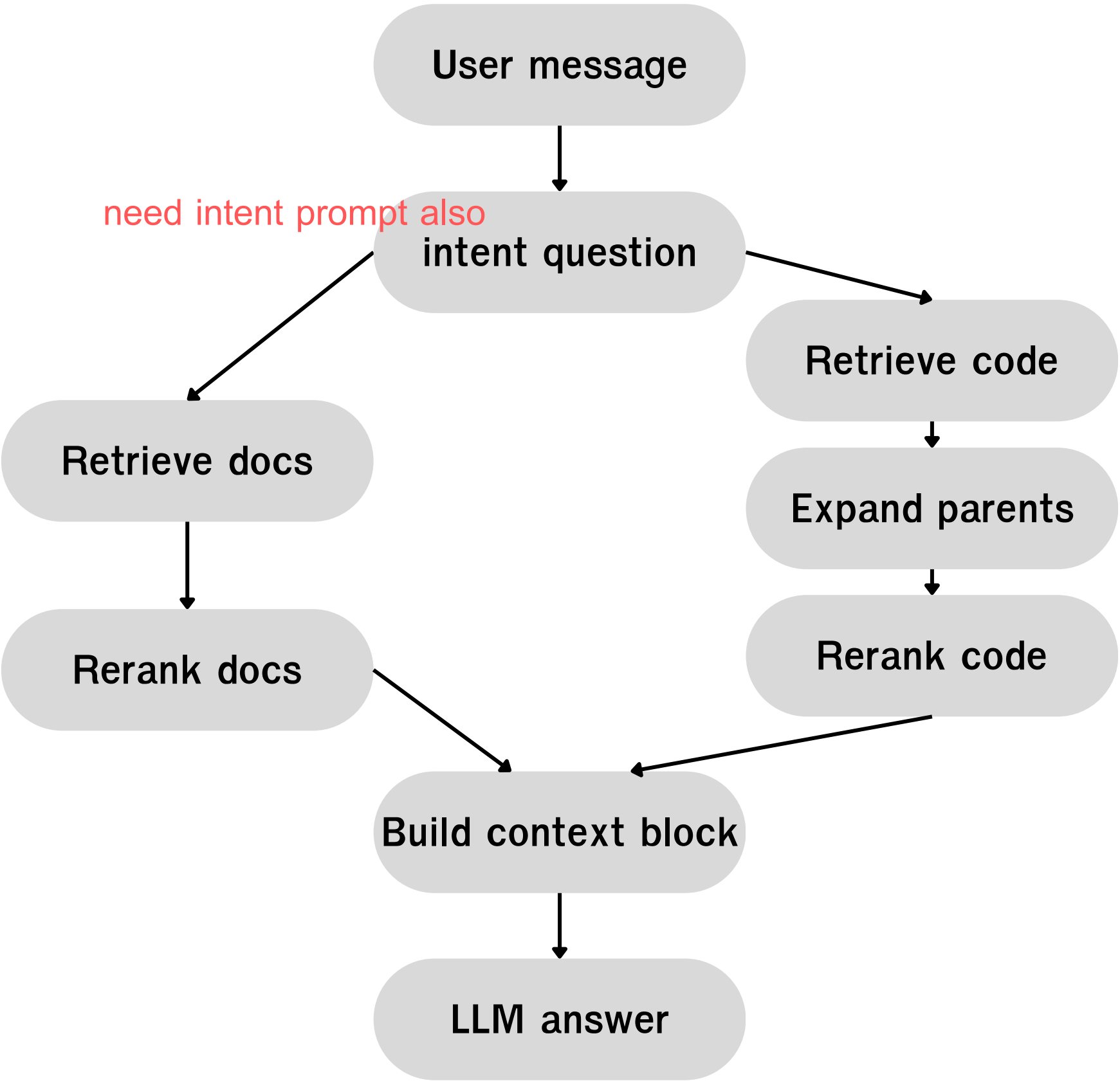
"How can I create a node using YAML EMC in EOL?"

query vector: [create: 0.4, node: 0.3, yaml: 0.2, eol: 0.1]

cosine similarity = $(0.4 \times 0.4) + (0.3 \times 0.2) + (0.2 \times 0.3) + (0.1 \times 0.1)$
 $= 0.16 + 0.06 + 0.06 + 0.01 = 0.29$

- Query 2 has higher similarity (0.29 vs 0.25) because its vector aligns better with the chunk
- create and node receive full weights instead of being diluted by studentId, courses, and attendance.

adding intent

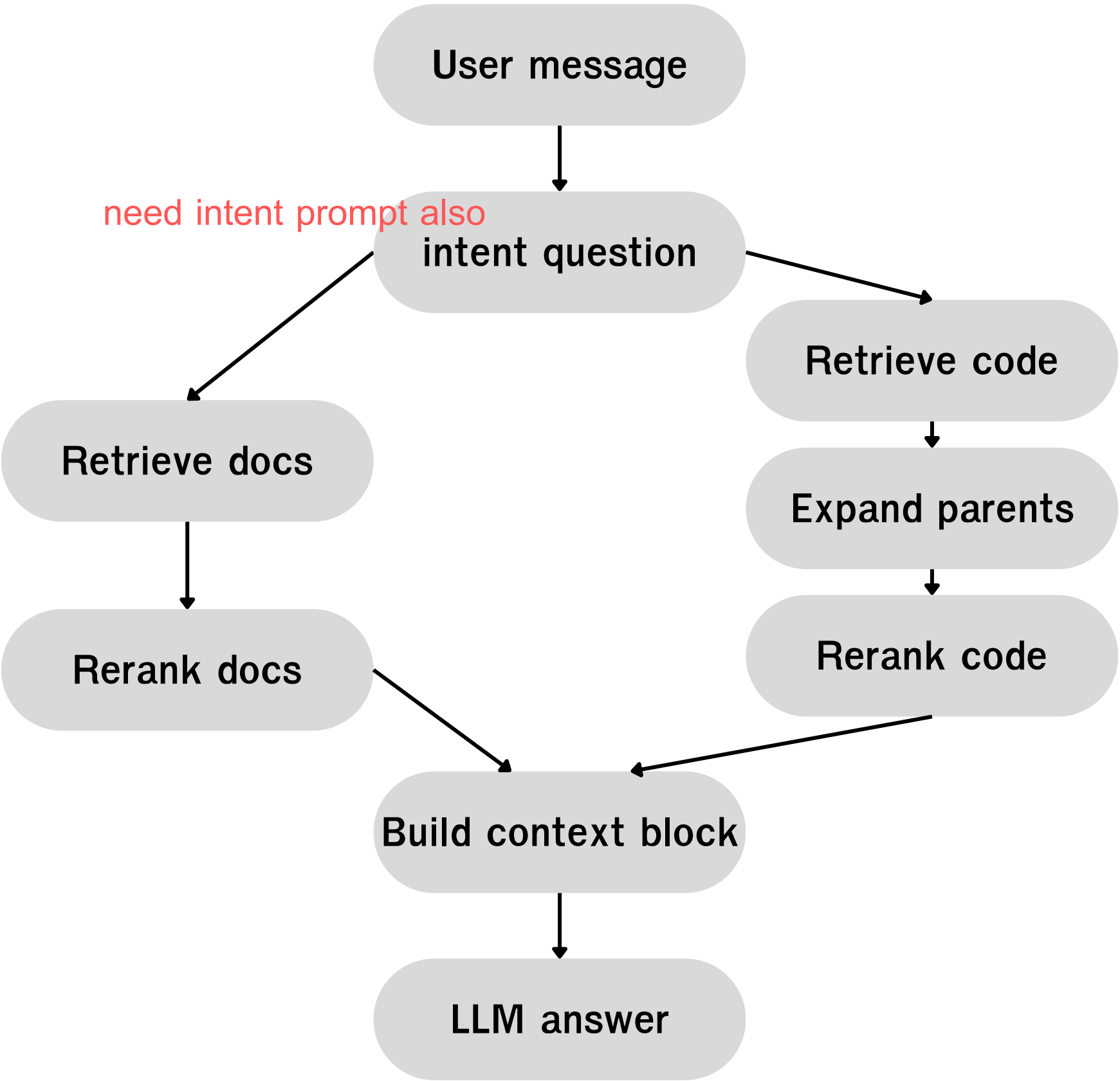


llm: gpt-4.1-mini

give all full code	give example	can run with no e	correct	complete	retrieval	faithfulness
yes	yes	yes	no	no	yes	medium
no	yes	yes	yes	no	yes	high
no	no	no	no	no	no	low
no	yes	yes	yes	no	yes	high
no	yes	yes	yes	no	yes	high
no	yes	yes	yes	no	yes	high
no	yes	no	no	no	yes	medium
no	yes	no	no	no	yes	medium
no	yes	yes	yes	no	yes	high
no	yes	no	no	no	yes	medium

- Get more related document → answer more correctly
- from case that can't retrieve → LLM intent question wrong → can't retrieve related document

adding intent



llm: gpt-5-mini

give all full code	give example	can run with no e	correct	complete	retrieval	faithfulness
yes	yes	yes	yes	yes	yes	high
yes	yes	yes	yes	yes	yes	high
yes	yes	yes	yes	yes	yes	high
yes	yes	yes	yes	no	yes	high
yes	yes	yes	yes	no	yes	high
yes	yes	yes	yes	no	yes	high
no	yes	no	no	no	yes	medium
no	yes	no	no	no	yes	medium
no	yes	yes	yes	no	yes	high
no	yes	yes	yes	no	yes	high

- intent correct question → can find related documents

run same case 3 times

	case 1	case 2	case 3	case 4	case 5	case 6	case 7	case 8	case 9	case 10
structure level	1	2	3	4	5	2	3	1	2	3
	40	90	80	120	160	90	150	80	110	120
give all full code	1	1	1	1	0	0.3333333333	1	1	1	1
give example	1	1	1	1	1	1	1	1	1	1
can run with no error	0.6666666667	1	1	0.6666666667	1	1	0	0.6666666667	1	0.6666666667
correct	0.6666666667	1	1	0.6666666667	0.3333333333	0.6666666667	0	0.6666666667	0.6666666667	0.6666666667
complete	0.6666666667	1	0.6666666667	0.6666666667	0	0.3333333333	0	0.6666666667	0.6666666667	0.6666666667
retrieval	1	1	1	1	1	1	1	1	1	1

for case mostly fail

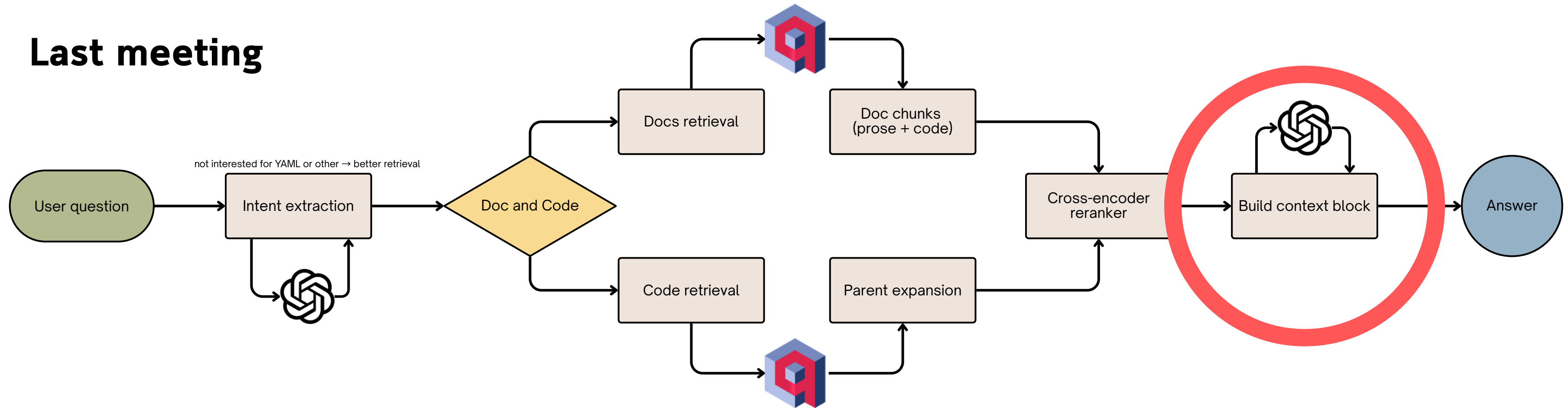
- having long line of the code → not generate full of code
- complex structure of YAML

for some easy and medium

- depends on LLM hallucunation
 - heading of code not complete
 - mapping(null)

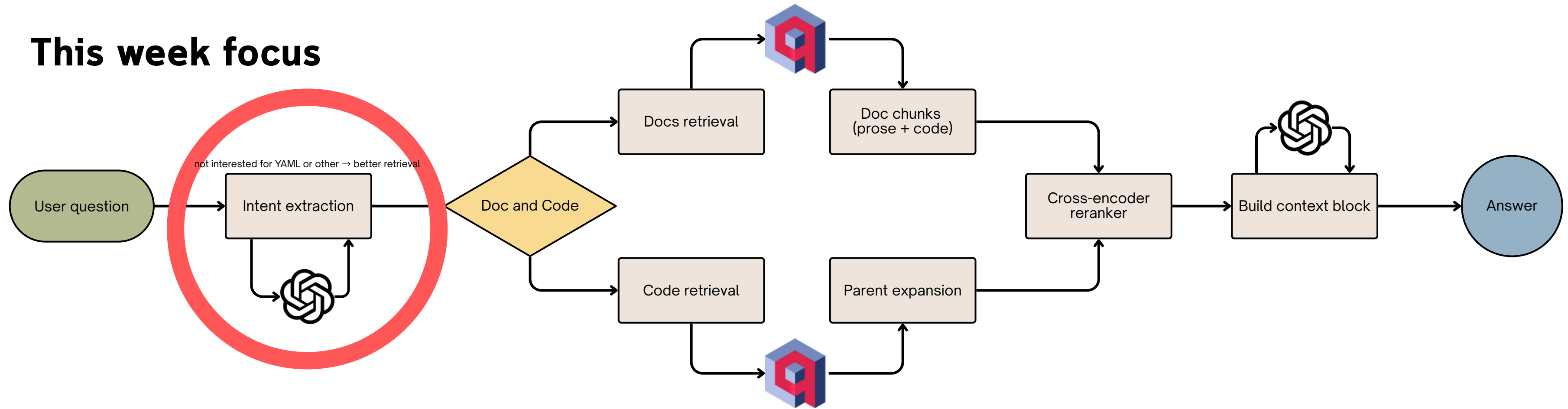
updated week 4

Last meeting



- Improved the answer prompt based on previously identified issues
- Found that the most critical factor affecting answer correctness is document retrieval
 - retrieving unrelated documents directly degrades answer quality

This week focus



- improve intent extraction part that can get better retrieve related document

```

INTENT_PROMPT = ChatPromptTemplate.from_template(
    """Extract the core Eclipse Epsilon technical question
from the user's message.
Ignore any sample data, YAML content, metamodel
definitions, or code snippets the user pasted.
Focus only on what they want to know or achieve in
Eclipse Epsilon terms.

Return the extracted question as the first line.
Then generate up to 2 additional search queries to retrieve
relevant documentation and code examples.
At least one of the additional queries should focus on the
Epsilon language syntax needed to accomplish the task.
Each query must use only terms that would naturally
appear in Eclipse Epsilon documentation — do not invent
API names or class names.
Return one query per line, nothing else.

Message:
{message}

Queries:"""
)

```

- improve intent extraction by generating multiple queries. The retrieval pool covers a broader range of relevant content

example

- How can I create YAML structures in EOL?
- EOL lists and maps syntax examples
- Epsilon EOL write text to file YAML serialization examples

from 10 cases

- 7 cases can run
- 6 cases can run with correct and complete code

error

- If it can't retrieve → can't answer
- invalid syntax
 - `portfolioNode.appendNode(new ScalarNode("id", "PF-UK-00771"));`

from retrieve

- related document get low score if compare other
- see ranking score is weird because of reranking model mismatch
 - now : cross-encoder/ms-marco-MiniLM-L-6-v2
 - change : BAAI/bge-reranker-v2-m3

1. [7.559] `epsilon_docs/eol.md`
Creating and Deleting Model Elements
2. [7.053] `epsilon_docs/eol.md`
Creating and Deleting Model Elements
3. [6.164] `epsilon_docs/articles_excel_index.md`
Creating Rows
4. [5.413] `epsilon_docs/articles_yaml-emc_index.md`
How can I delete nodes?
5. [4.598] `epsilon_docs/articles_yaml-emc_index.md`
Scripting YAML documents using Epsilon
6. [4.580] `epsilon_docs/articles_yaml-emc_index.md`
Querying/modifying YAML documents in EOL
7. [3.822] `epsilon_docs/articles_yaml-emc_index.md`
How can I create a YAML document from scratch?
8. [3.737] `epsilon_docs/articles_xsd-xml_index.md`
How do I create an element and add it to an existing element?

from changing reranking model

- get 100% for retrieve related document
- related document get better score

intent prompt and answer prompt not change

1. [0.977] epsilon_docs/articles_csv-emc_index.md
How do I create an element?
2. [0.975] epsilon_docs/eol.md
Creating and Deleting Model Elements
3. [0.892] epsilon_docs/articles_excel_index.md
Creating Rows
4. [0.890] epsilon_docs/articles_yaml-emc_index.md
How can I create a YAML document from scratch?
5. [0.828] epsilon_docs/eol.md
Creating and Deleting Model Elements
6. [0.270] epsilon_docs/articles_yaml-emc_index.md
Scripting YAML documents using Epsilon
7. [0.243] epsilon_docs/articles_yaml-emc_index.md
Querying/modifying YAML documents in EOL

from changing reranking model

- get 100% for retrieve related document
- related document get better score

from intent extraction

- Return the extracted question as the first line.
 - this line lead to don't get actual user question
example
user: How can I create YAML structures in EOL?
intent: How can I create YAML structures (nested maps and lists) using EOL?
 - → related document score lower than it should be

1. [0.977] epsilon_docs/articles_csv-emc_index.md
How do I create an element?
2. [0.975] epsilon_docs/eol.md
Creating and Deleting Model Elements
3. [0.892] epsilon_docs/articles_excel_index.md
Creating Rows
4. [0.890] epsilon_docs/articles_yaml-emc_index.md
How can I create a YAML document from scratch?
5. [0.828] epsilon_docs/eol.md
Creating and Deleting Model Elements
6. [0.270] epsilon_docs/articles_yaml-emc_index.md
Scripting YAML documents using Epsilon
7. [0.243] epsilon_docs/articles_yaml-emc_index.md
Querying/modifying YAML documents in EOL

from changing reranking model

- get 100% for retrieve related document
- related document get better score

from intent extraction

- Return the extracted question as the first line.
 - this line lead to don't get actual user question
example
user: How can I create YAML structures in EOL?
intent: How can I create YAML structures (nested maps and lists) using EOL?
 - → related document score lower than it should be
- add “, with only sample data and code snippets removed.”
 - retrieve score is better

1. [0.977] epsilon_docs/articles_csv-emc_index.md
How do I create an element?
2. [0.975] epsilon_docs/eol.md
Creating and Deleting Model Elements
3. [0.892] epsilon_docs/articles_excel_index.md
Creating Rows
4. [0.890] epsilon_docs/articles_yaml-emc_index.md
How can I create a YAML document from scratch?
5. [0.828] epsilon_docs/eol.md
Creating and Deleting Model Elements
6. [0.270] epsilon_docs/articles_yaml-emc_index.md
Scripting YAML documents using Epsilon
7. [0.243] epsilon_docs/articles_yaml-emc_index.md
Querying/modifying YAML documents in EOL

- now can retrieve 100% (all cases)
- but generated answer still wrong some case

- over generated

```
- name: Spice Level
  required: true
  options:
  - option: {
    }
    label: Mild
    priceDelta: 0
```

```
- name: Spice Level
  required: true
  options:
    label: Mild
    priceDelta: 0
```

- still have invalid syntax

```
patient.appendNode(new ScalarNode("patientId", "PT-00291837"));
```

```
patient.value.appendNode(new ScalarNode("patientId", "PT-00291837"));
```

- missing starting or incomplete

```
YAMLDoc.setRootAsList();
var root = YAMLDoc.getRoot();
root.value.addRows(1);
var doc = root.value.at(0);
```

- now can retrieve 100% (all cases)
- but generated answer still wrong some case

- append to the root directly (need to append to row)

root.appendChild(trip);

- not have a list as a root → give map

booking:
reference: BKNG-TH7X92
createdAt: '2024-05-01T13:22:00Z'
status: confirmed

- booking:
reference: BKNG-TH7X92
createdAt: "2024-05-01T13:22:00Z"
status: confirmed

- generating some strange things.

itemsS01.value.at(itemsS01.value.at(0).value.size()-1).

- no wrapper

- term: Spring Term 2024
issueDate: '2024-04-05'
school:
name: Greenfield Primary School

- report:
term: Spring Term 2024
issueDate: "2024-04-05"
school:
name: Greenfield Primary School

updated week 5

- create testcase

Tier 1 - each axis

- Node Composition
- Topological Structure (structure)
- Schema Consistency
- Syntax-Level
- Document-Level

Tier 2 - combine 2 axes

Tier 3 - mixed

case	type	Lines	root	records(if	Depth	Category	Isolated Axis
1	1	6	map		2	Node Composition	Mixed scalar value types (string, integer, float, boolean, null)
2	1	5	map		2	Node Composition	String values that resemble numeric or date types
3	1	3	map		3	Node Composition	Low mapping fan-out (2 keys)
4	1	11	map		3	Node Composition	High mapping fan-out (10 keys)
5	1	5	map		2	Node Composition	Homogeneous list of scalar values (Sequence{} assigned directly)
6	1	10	map		3	Node Composition	List of mappings (addRow + appendNode)
7	1	4	map		3	Topological Structure (structure)	Shallow mapping depth
8	1	8	map		6	Topological Structure	Medium mapping depth
9	1	25	map		10	Topological Structure	Maximum mapping depth with no lists
10	1	9	map		3	Topological Structure	Shallow list nesting
11	1	32	list	1	6	Topological Structure	Maximum list nesting while keeping mapping depth shallow (counterpart to Case 09)
12	1	9	map		3	Topological Structure	Wide but shallow hierarchy (counterpart to the narrow and deep structure in Case 09)
13	1	13	map		3	Schema Consistency	List whose entries share an identical key set
14	1	28	list	4	3	Schema Consistency	List whose entries have completely different mapping schemas
15	1	9	map		3	Schema Consistency	Combination of null scalar, null object, empty list ({}), and empty mapping ({})
16	1	4	map		3	Syntax-Level	Entire document written in flow style ({} / [])
17	1	9	map		3	Syntax-Level	Mixed naming conventions (camelCase and snake_case) within the same document
18	1	6	map		2	Node Composition	Heterogeneous scalar list containing multiple data types
19	1	49	list	10	3	Document-Level	Large multi-root document with 10 uniformly shallow records

- expand parent-child (eg. H1-H3) node for documents
 - can retrieve better but still effect by ranking
- deduplicated documents from 3 set of relevent intent
 - How can I create a node using the YAML EMC in EOL? → 3
 - YAML EMC create model element EOL → 3
 - EOL syntax for creating new model elements with YAML EMC → 3
 → deduplicated → reranking
- change user question form to test - see limitation
 - How can I create a node using the YAML EMC in EOL?
 - How can I create YAML structures in EOL?
- LLM hallucination when generated map and list
 - if can retrieve all releveant documents
 - if YAML is a list that has 1 record → it change to map
 - if can't retrieve set root document
 - generated with list
- edit answer prompt to be more generic

- expand parent-child (eg. H1-H3) node for documents
 - can retrieve better but still effect by ranking

Q: How can I create a node using the YAML EMC in EOL?

Scripting YAML documents using Epsilon

In this article we demonstrate how you can create, query and modify YAML documents in Epsilon programs using the YAML driver. All the examples in this article demonstrate using EOL to script YAML documents. However, it's worth stressing that YAML documents are supported throughout Epsilon. Therefore, you can use Epsilon to (cross-)validate, transform (to other models – XML, CSV or EMF-based, or to text), compare and merge your YAML documents.

How do I set the root node of a YAML document?

The `root` property can be used for setting the root node of a YAML document. The root node of a YAML document can either be a mapping node or a list node. The `setRootAsMap()` function can be used for setting the root node into a mapping node. The `setRootAsList()` function can be used for setting the root node into a list node.

```
// Set the root node into a mapping node
YAMLDoc.setRootAsMap();

// Set the root node into a list node
YAMLDoc.setRootAsList();

// Set the root node into a mapping node
YAMLDoc.root.value = new MappingNode();

// Set the root node into a list node with 3 empty entries
YAMLDoc.root.value = new ListNode("", 3);
```

How can I create a YAML document from scratch?

The following code can be used for creating the `employees.yaml` file programmatically.

```
YAMLDoc.setRootAsList();
var root = YAMLDoc.getRoot();
root.value.addRows(2);
var rootFirstRow = root.value.at(0);
var rootSecondRow = root.value.at(1);

var addressNode1 = new MappingNode("address");
var countryNode1 = new ScalarNode("country", "England");
var residentNode = new ScalarNode("resident", true);
var postcodeNode = new ScalarNode("postcode", "M1 1AG");
addressNode1.value.appendNode(countryNode1);
addressNode1.value.appendNode(residentNode);
addressNode1.value.appendNode(postcodeNode);
```

- change user question form to test - see limitation generic
 - How can I create a node using the YAML EMC in EOL? (not provide)
 - How can I create YAML structures in EOL?

How can I start building a YAML model from scratch in EOL?

specific

How can I create programmatically the YAML document I provided as input, from scratch using the EMC YAML Driver?

How can I create programmatically the YAML document I provided as input, from scratch using the EMC YAML Driver? First, you should figure out what is the root node, either a list node and mapping node, and change the code correspondingly

1. [0.890]
 epsilon_docs/articles
 _yaml-emc_index.md
 (heading='How can I
 create a YAML
 document from
 scratch?',
 parent_heading='Que
 rying/modifying
 YAML documents in
 EOL')

How can I
 create a YAML
 document from
 scratch?
 Example: create a
 YAML document from
 scratch by building
 and appending
 MappingNode,
 ScalarNode, and
 ListNode elements.
 Keywords:
 setRootAsList,
 getRoot

- LLM hallucination when generated map and list
 - if can retrieve all relevant documents
 - if YAML is a list that has 1 record → it change to map
 - if can't retrieve set root document
 - generated with list

updated week 6 (folder10)

YAML

<https://spacelift.io/blog/yaml#what-is-yaml>

A YAML format primarily uses three node types:

1. Maps/Dictionaries (YAML calls it mapping)
2. Arrays/Lists (YAML calls them sequences)
3. Literals (Strings, numbers, boolean, etc.)

Advanced YAML - not included

- Documents
- Schemas and Tags
- Anchors (&) and aliases (*)

User question

- How can I create programmatically the YAML document I provided as input, from scratch using the EMC YAML Driver?

result:

- From 47 cases
 - can run 34 cases
 - 30 cases generated wrong root
 - some case generated setRootAsMapping which is not having this syntax

User question

- How can I create programmatically the YAML document I provided as input, from scratch using the EMC YAML Driver? **First, you should figure out what is the root node, either a list node and mapping node, and change the code correspondingly.**

result:

- From 47 cases
 - can run 28 cases
 - 6 cases generated wrong root
 - but see use invalid syntax to append to map/list
 - map : root.value.append
 - list : root.append

User question

- How can I create programmatically the YAML document I provided as input, from scratch using the EMC YAML Driver? First, you should figure out what is the root node, either a list node and mapping node, and change the code correspondingly. **Make sure to use the correct method to append nodes for mapping nodes and list node slots.**

result:

- From 47 cases
 - can run 31 cases
 - still have invalid syntax to append to map/list
 - if it is a list that have only 1 record → generated as a map

User question

- How can I create programmatically the YAML document I provided as input, from scratch using the EMC YAML Driver? First, you should figure out what is the root node, either a list node and mapping node, and change the code correspondingly. Make sure to use the correct method to append nodes – use **.value.appendNode()** for mapping nodes and **.appendNode()** for list node slots.

result:

- From 47 cases
 - can run 37 cases
 - invalid syntax to append to map/list have been fixed
 - if it is a list that have only 1 record → generated as a map
 - if can't retrieve how to set root as a map → hallucinate to generated setRootAsMapping

General Question (EOL, EGL, EGX, EVL, ETL, and YAML EMC)

Use old prompt

- can answer correct
- but generate with full runnable example even conceptual question

fixed prompt - “Match the scope of your answer to the user's request.”

- examples are still included, but they have been made more concise in several cases
- found that if can't retrieve relevant document → it will answer wrong

updated week 7 (folder11)

Choose 5 testcases which have error most

- 1.Low mapping fan-out (2 keys)
- 2.Map which have list whose entries share an identical key set
- 3.Map root mixing a flat section with a list section, where list items don't all follow the same structure (depth 6)
- 4.List root with multiple records sharing an identical, deeply-nested schema
- 5.List root wrapping a single deeply-nested map (depth ~10, mixed list/map/scalar)

case	1 (map)	2 (map)	3 (map)	4 (list 2 records)	5 (list 1 record)	sum
GPT-5-mini no clue (base-line)	X (no set root doc)	X (no set root doc)	X (no set root doc)	✓ (no set root doc)	X (no set root doc)	1
GPT-5-mini (last week) with clue in question	✓	✓	✓	✓	X	4
GPT-5-mini with clue in prompt	✓ (no set root doc)	✓ (no set root doc)	✓ (no set root doc)	✓ (no set root doc)	X (no set root doc)	4
GPT-5 no clue	✓	X (no set root doc)	✓	✓	✓ (no set root doc)	4
GPT-5 (last week) with clue in question	✓	✓	✓	✓	✓	5
GPT-5 with clue in prompt	✓	✓	✓ (no set root doc)	✓ (no set root doc)	✓ (no set root doc)	5

From baseline

- Can't retrieve how to set root document
 - → hallucination create syntax / generated list as a root
- The one that correct
 - Having root as a list → use only how to create YAML is enough

Moving clue context to prompt

- Can't retrieve how to set root document
- But use a clue from a prompt to generate
- Still having issue for 1 record which have list as a root

Changing LLM model to GPT-5

- Have more chance that can retrieve how to set root document because intent question is better than using GPT-5-mini
 - Cases that can retrieve that document → generated correct

Using GPT-5 with giving a clue in the prompt

- Can answer correct for all cases (5 cases)
- Interesting that the List root wrapping a single cases can answer correct
 - → meaning that GPT-5 can know more about this YAML structure

case (no clue)	1 (map)	2 (map)	3 (map)	4 (list 2 records)	5 (list 1 record)	sum
GPT-5-mini	X	X	X	✓	X	1
GPT-5	✓	✓	✓	✓	✓	4
claude-opus-4.8	X	X	X	✓	X	1
claude-sonnet-5	✓	✓	✓	✓	✓	5
gemini-3.1-pro	✓	✓	X	✓	✓	4
gemini-3.5-flash	✓	✓	✓	✓	✓	5
qwen3.6-flash	✓	✓	✓	✓	✓	5

Normal Question Prompt - depends on what we want

Component

- Directive
- Examples
- Output Formatting
- Style Instructions
- Role (Persona)
- Additional Information

Remark from paper:

- Not use “an expert assistant specialised” → hallucination and not answer that don’t know
- Try to not using “context”

Link Ref: <https://arxiv.org/abs/2406.06608>

Normal Question Prompt

- baseline
- Role: changing role → expert persona
- Output Formatting: focus only prose, giving code if user want
- Style: be concise, not too expand
- Examples: ask to flag it explicitly
- Directive: scope-first (instruction for code part only)

Choose 5 testcase which have weak until partial evidence (focus weak)

Role Told the AI "you're an expert"	Worse, AI became overconfident, started guessing without flagging it as a guess
Output format Instructed it to answer in prose, not excessive code	Answers got shorter, which was good, but still overconfident.
Style Instructed it to "match answer length to how much evidence actually exists"	Better. It started saying "I don't know" again when it really didn't know
Examples Gave an example phrase for flagging uncertainty ("this isn't confirmed...")	Good at showing where info came from
Directive Instructed the AI to break the question into sub-parts before answering	Good. It's better to answer by identifying which parts you're unsure about, rather than saying you understand everything and just guessing.

LLM as a judge

give

- question
- context
- answer

and tell it to

1. What does the context actually say that's relevant to the question?
2. Does the answer match what the context says, or does it add/contradict anything?
3. Does the answer fully address the question?

Then give a score from 1 to 10.

ref: <https://arxiv.org/abs/2306.05685>

<https://www.confident-ai.com/blog/why-llm-as-a-judge-is-the-best-llm-evaluation-method>

updated week 8 (folder12)

Result compare LLM Model

Using LLM-as-a-judge to give score 1-10 about generated answer

- Answer Correctness: semantic agreement with the ground truth.
- Faithfulness: every factual, API, syntax, and code claim is supported by retrieved context.
- Answer Relevancy: directly addresses the question without irrelevant material.
- Completeness: covers every explicit requirement and all essential reference points.

METHODOLOGY

1.index design

- chunking strategy
- using docs+code or only docs

2.query understanding

- intent question + query expanding

3.retrieve design

- retrieve strategy
- sibling expanding
- reranking

4.generation answer design

- answer prompt (last week)
- instruction placement (last week)
- LLM model choice (this week)

EVALUATION METRICS

1.index design

- chunking strategy → no experiment
- using docs+code or only docs → compare experiment

2.query understanding → compare

- intent question + query expanding

3.retrieve design → Recall@K, Precision@K, nDCG, MAP

- retrieve strategy
- sibling expanding
- reranking

4.generation answer design

- answer prompt (last week) → no experiment, just inspired
- instruction placement (last week) → mini-experiment
- LLM model choice (this week) → LLM as a judge with Recall@K, Precision@K, nDCG, MAP

Reference

- nDCG(Normalized Discounted Cumulative Gain)
- MAP(Mean Average Precision)